

IIT KHARAGPUR

Department of Civil Engineering

A Project Report on

STRUCTURAL ANALYSIS OF G+5 RCC BUILDING USING ETABS

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1. Introduction

Reinforced cement concrete (RCC) framed construction remains the most widely adopted structural system for multi-storey residential and institutional buildings in India, owing to its constructional flexibility, established design codes, and ready availability of materials. As buildings gain height and plots become more constrained, structural analysis becomes essential to confirm that the proposed frame can safely resist both gravity loads and lateral demands, particularly seismic forces, which govern the design of most Indian buildings located in seismically active regions.

This report presents the structural analysis of a Ground plus five storey (G+5) reinforced concrete framed building, carried out using ETABS, a widely used industry-standard software for the analysis and design of building structures. The analysis covers gravity load computation, equivalent static seismic analysis as per IS 1893 (Part 1):2016, and interpretation of key structural response parameters such as base shear, storey drift, storey displacement, and torsional behaviour. The study is intended to demonstrate a complete, code-compliant structural analysis workflow suitable for academic evaluation and as a foundation for subsequent detailed design.

2. Objectives of the Project

- To idealise and model a G+5 RCC framed building in ETABS, incorporating realistic geometry, member sizes, and support conditions.
- To compute dead, live, and seismic loads acting on the building in accordance with IS 875 (Parts 1 and 2) and IS 1893 (Part 1):2016.
- To perform modal analysis and determine the fundamental time period and mode shapes of the building.
- To evaluate the base shear, storey displacement, storey drift, and storey stiffness of the building under seismic loading in both principal directions.
- To check the building for torsional irregularity as defined under IS 1893:2016, and identify load combinations requiring design attention.
- To draw conclusions on the overall structural adequacy of the building and recommend measures for detailed design and future study.

3. Building Description and Project Data

The building under study is a residential G+5 RCC framed structure with a rectangular plan of 12 m × 4 m. The longer (X) direction is divided into three bays of 4 m each, while the shorter (Y) direction comprises a single 4 m bay. The building comprises a ground storey and five upper storeys, each of 3.2 m height, giving a total height of 19.8 m above ground level.

Parameter	Value
Type of structure	G+5 RCC framed building
Plan dimensions	12 m × 4 m
Number of bays (X-direction)	3 bays @ 4 m c/c
Number of bays (Y-direction)	1 bay @ 4 m
Typical storey height	3.2 m
Total building height	19.8 m
Seismic zone	Zone II

Table 3.1: Building configuration and project data

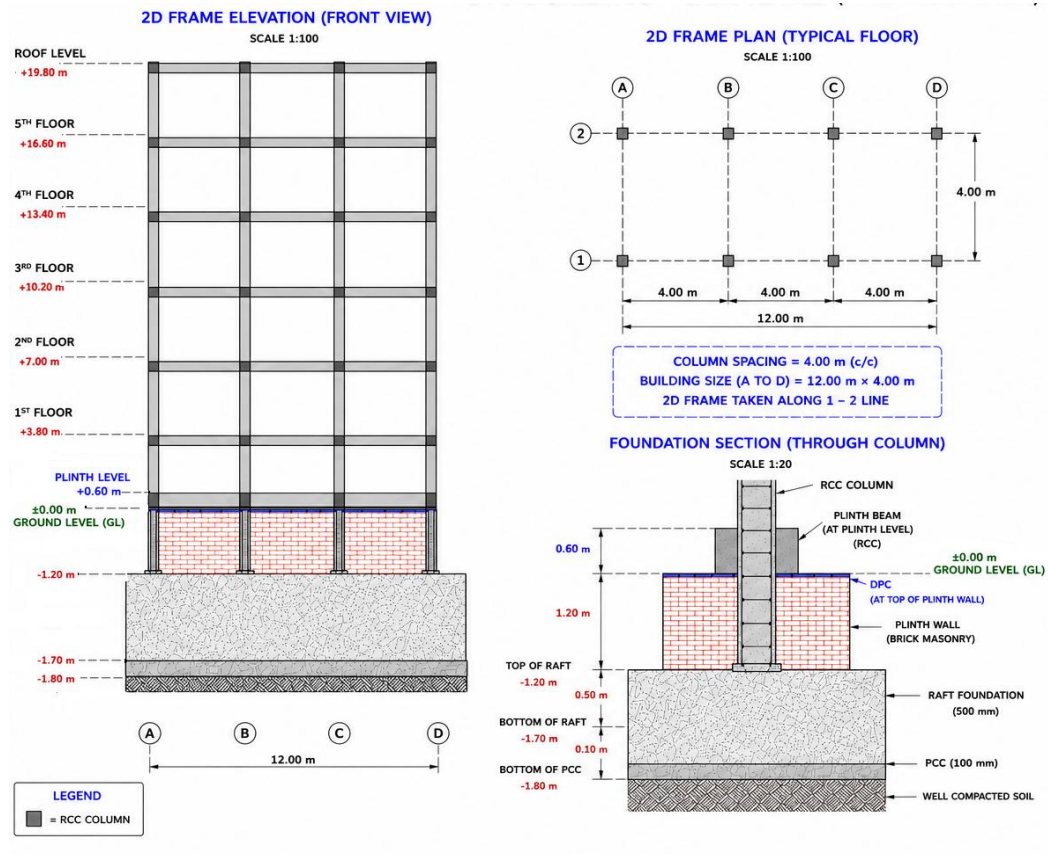


Figure 3.1: Frame elevation, typical floor plan, and foundation section

4. Material Specifications

Material	Grade / Specification
Concrete (columns, beams, slabs, raft)	M25
Reinforcing steel	Fe 500
PCC (below raft)	M10 (1:3:6)
Plinth wall masonry	Brick masonry, 1:6 cement mortar
Unit weight of RCC (for dead load computation)	25 kN/m ³

Table 4.1: Material specifications adopted

5. Structural Member Details

Member	Size
Column	450 mm × 450 mm
Beam (typical floor)	300 mm × 500 mm
Plinth beam	300 mm × 500 mm, at plinth level
Slab (assumed for load transfer)	125 mm

Table 5.1: Structural member sizes

Uniform member sizes and storey heights were maintained throughout the height of the building, consistent with standard practice for a regular residential frame of this scale.

6. Foundation Details

The building is supported on a raft foundation, selected for its capacity to distribute column loads uniformly over the founding strata and to control differential settlement across the closely spaced column grid.

Parameter	Value
Foundation type	Raft foundation
Raft thickness	500 mm
PCC thickness (below raft)	100 mm, M10 (1:3:6)
Top of raft	–1.20 m
Bottom of raft	–1.70 m
Bottom of PCC	–1.80 m
Founding stratum	Well-compacted soil

Table 6.1: Foundation details

7. Design Codes Used

Code	Application
IS 456:2000	Plain and Reinforced Concrete — general design provisions
IS 875 (Part 1):1987	Dead loads
IS 875 (Part 2):1987	Imposed (live) loads
IS 1893 (Part 1):2016	Earthquake resistant design of buildings
IS 13920:2016	Ductile detailing of RC structures for seismic zones

Table 7.1: Design codes referred to in this project

8. Load Calculations

8.1 Dead Load

Dead loads were computed from member self-weights using a unit weight of 25 kN/m³ for reinforced concrete, in accordance with IS 875 (Part 1). Beam self-weight was computed as $0.30 \times 0.50 \times 25 \times 4 = 15$ kN per beam, and column self-weight as $0.45 \times 0.45 \times 25 \times 3.2 = 16.2$ kN per storey height. Summed across the representative frame, these give the totals below.

Component	Weight (kN)
Total beam self-weight	270.00
Total column self-weight	328.05
Total dead load	598.05

Table 8.1: Dead load summary

8.2 Live Load

Live loads were assigned as per IS 875 (Part 2): 2 kN/m² for residential floors and 1.5 kN/m² for the (accessible) roof. Applied over the 48 m² plan area across five typical floors and the roof, this gives a total live load of 552 kN (480 kN from floors and 72 kN from the roof), adopted as the design live load for this project.

8.3 Seismic Load

Seismic loads were derived using the equivalent static lateral force method of IS 1893 (Part 1):2016 for Seismic Zone II, applicable given the building's regular configuration and height. The design horizontal seismic coefficient and resulting storey forces are discussed in Section 9, and the corresponding base shear results are presented in Section 11.2.

9. Seismic Analysis Parameters

Parameter	Value	Basis
Seismic zone	II	Low seismicity region
Zone factor, Z	0.10	IS 1893:2016, Table 3
Importance factor, I	1.0	Residential occupancy
Response reduction factor, R	5	Special moment resisting frame (SMRF)
Soil type	Medium soil	Assumed
Fundamental time period, T	0.611 s	ETABS modal analysis
Design horizontal seismic coefficient, Ah	≈ 0.0223	Computed as below

Table 9.1: Seismic design parameters

The design horizontal seismic coefficient was computed using $A_h = (Z/2) \times (I/R) \times (S_a/g)$. With $T = 0.611$ s (from modal analysis) on medium soil, $S_a/g = 1.36/T \approx 2.23$. Substituting $Z = 0.10$, $I = 1.0$ and $R = 5$ gives $A_h = (0.10/2) \times (1.0/5) \times 2.23 \approx 0.0223$, which was used to derive the equivalent static seismic loads (EQX and EQY) applied to the ETABS model.

10. Load Combinations Used

The following fourteen load combinations, based on IS 456:2000 and IS 1893 (Part 1):2016, were defined in ETABS for the strength and serviceability checks discussed in this report:

- 1.5(DL + LL)
- 1.2(DL + LL ± EQX)
- 1.2(DL + LL ± EQY)
- 1.5(DL ± EQX)
- 1.5(DL ± EQY)
- 0.9DL ± 1.5EQX
- 0.9DL ± 1.5EQY

11. Analysis Results and Discussion

11.1 Modal Analysis

Mode	Period (s)	Predominant Direction
1	0.611	Y (translational)
2	0.504	Torsional
3	0.427	X (translational)

Table 11.1: Fundamental modal periods

Cumulative modal mass participation reached 100% in both principal directions by the 20th mode, satisfying the IS 1893:2016 requirement of capturing at least 90% of the seismic mass, confirming that the modal analysis adequately represents the dynamic behaviour of the building.

11.2 Base Shear

Direction	Base Shear (tonf)
EQX	62.33
EQY	61.65

Table 11.2: Design base shear

The base shears obtained in the X and Y directions are comparable in magnitude, reflecting the common seismic weight and design parameters applied in both directions.

11.3 Storey Displacement

The maximum lateral displacement at roof level was found to be 48.6 mm under the EQY load case, occurring at the roof diaphragm centre of mass. Displacements were found to increase progressively with height, consistent with the expected cantilever-type deformation profile of a regular moment-resisting frame.

11.4 Storey Drift

Direction	Maximum Storey Drift Ratio	IS 1893:2016 Limit
X	0.0018	0.004
Y	0.0034	0.004

Table 11.3: Storey drift check

Both directions satisfy the IS 1893:2016 drift limit of 0.004 times the storey height, confirming that the building meets the code's serviceability requirement for lateral drift control.

11.5 Storey Stiffness

Storey lateral stiffness was found to increase progressively from the roof towards the ground storey in both directions, which is the expected trend for a regular frame with uniform member sizes and storey heights. No abrupt reduction in stiffness was observed at any level, indicating a regular vertical stiffness distribution over the height of the building.

11.6 Torsional Irregularity Check

The diaphragm displacement ratio — the ratio of maximum to average displacement at each floor diaphragm under the design eccentric seismic combinations — was examined at all storeys as per IS 1893:2016. This ratio remained within acceptable limits for the majority of the fourteen load combinations considered. Two combinations (representing counteractive seismic load cases) showed a diaphragm displacement ratio exceeding the code threshold at the first storey, indicating a torsional irregularity that will need to be addressed during detailed design, as summarised in the Conclusions.

12. Observations

- The building exhibits comparable base shear demand in both principal directions, consistent with the near-symmetric application of seismic weight and design parameters.
- Storey displacements and drifts increase steadily with height, following the expected deformation pattern of a regular moment-resisting frame.

- Storey stiffness increases progressively towards the base, confirming a regular vertical stiffness distribution with no abrupt irregularities.
- Modal mass participation reaching 100% by the 20th mode confirms that the dynamic analysis adequately captures the building's seismic response.
- The Y-direction (narrower plan dimension) governs both drift and torsional behaviour, warranting attention during detailed design.

13. Conclusions

- The G+5 RCC framed building, with a 12 m × 4 m plan and 19.8 m total height, was successfully modelled and analysed in ETABS for gravity and equivalent static seismic loads under Zone II conditions as per IS 456:2000 and IS 1893 (Part 1):2016.
- The total dead load and live load on the representative frame were found to be approximately 598.05 kN and 552 kN respectively, consistent with hand-calculated values based on IS 875 (Parts 1 and 2).
- The design horizontal seismic coefficient, A_h , was computed as approximately 0.0223 for Seismic Zone II, using $Z = 0.10$, $I = 1.0$ and $R = 5$.
- The fundamental time period of the building was found to be 0.611 s, with base shears of 62.33 tonf (EQX) and 61.65 tonf (EQY).
- Maximum storey drift ratios of 0.0018 (X) and 0.0034 (Y) were both within the IS 1893:2016 limit of 0.004, confirming satisfactory drift control.
- Storey stiffness was found to increase regularly towards the base in both directions, with no soft-storey irregularity observed.
- Of the fourteen load combinations examined, two combinations were found to exhibit extreme torsional irregularity at the first storey, exceeding the IS 1893:2016 diaphragm displacement ratio limit; this will need to be specifically addressed — through stiffness enhancement or plan modification — before the design is finalised.
- Overall, the building demonstrates satisfactory structural performance under the considered load cases, subject to resolution of the identified torsional irregularity.

14. Recommendations / Future Scope

- Address the torsional irregularity identified in the two governing load combinations, through measures such as introducing a shear wall or bracing along the narrower plan direction, before proceeding to final design.
- Extend the study to a complete three-dimensional design-level analysis incorporating all gravity and lateral load-resisting frames.
- Carry out response spectrum analysis for a more refined estimate of seismic demand, particularly for the direction found to be more sensitive to drift and torsion.
- Proceed to detailed limit-state design and reinforcement detailing of beams, columns, and the raft foundation using IS 456:2000 and IS 13920:2016.
- Re-verify the torsional irregularity and drift results after implementing the recommended stiffening measures, prior to finalising construction drawings.